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Impact of land use conversion patterns on vertical hydrological connectivity in intensive orchards

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ABSTRACT

Vertical hydrological connectivity (VHC), plays a crucial role in regulating soil water and nutrient cycles, thereby contributing to the sustainability of vegetation restoration. The impacts of anthropic factors, particularly land use changes, on vertical hydrology have gained increasing attention. However, the mechanisms underlying the effects of land use conversion patterns on VHC remain insufficiently understood. To address this knowledge gap, this study investigated the VHC of orchard soils converted from paddy fields (P-O sites) and drylands (D-O sites) using dye-tracer experiments. VHC was quantified using key indicators, including dye coverage (DC), maximum dyed depth (MMD), length index (L_i), depth of diffusion area (U_{niFr}), and percentage of preferential pathways (P_{F-fr}). It showed that compared with P-O site, D-O sites exhibited significantly higher sand and gravel content, along with greater non-capillary porosity. Additionally, a relatively uniform and vertically distributed root system was observed at D-O sites. Thus, DC at D-O sites was 63.87 % higher than that at P-O sites, indicating stronger hydrological connectivity. Furthermore, U_{niFr} at D-O sites was 12 times higher than that at P-O sites, while P_{F-fr} exhibited the opposite trend, suggesting a more homogeneous infiltration pattern at D-O sites. However, no significant differences were observed in MMD and L_i. At P-O sites, dye coverage fluctuated with depth, peaking at 10–20 cm, whereas at D-O sites, a consistent decreasing trend was observed. The stratification ratio (SR) of soil properties were identified as the most critical factor influencing VHC, explaining 70 % of the variation. Specifically, soil porosity (r = -0.60) had a direct impact on VHC, while soil texture (r = 0.80) and root distribution (r = -0.39) primarily influenced VHC indirectly through their effects on soil porosity. These findings suggest that the presence of a hydrological barrier layer in orchards converted from paddy fields restricts hydrological connectivity. To mitigate this limitation, appropriate tillage practices, such as deep plowing, should be implemented to disrupt the impermeable layer and enhance VHC.

1. Introduction

Hydrological connectivity plays a crucial role in environmental processes and ecosystem functions by governing the transport of substances during the hydrologic cycle (Conte and Ferro, 2018). It exhibits three primary dimensions: longitudinal, lateral, and vertical (Merenlender and Matella, 2013). Longitudinal connectivity describes water movement along tidal channels, while lateral connectivity refers to flow perpendicular to the channel; both are largely controlled by topography and river or channel networks (Wohl and Beckman, 2014; Granados et al., 2020). Vertical connectivity, on the other hand, defines the extent to which solutes and insoluble organic matter can migrate through soil, driven by water flow in the vertical direction (Merenlender and Matella, 2013; Covino, 2017; Dai et al., 2020). This vertical hydrological connectivity governs water movement pathways and velocities within soil, thereby influencing groundwater recharge, nutrient transport, and biological processes (Wu et al., 2021). It is essential for maintaining regional water balance, regulating runoff, facilitating nutrient cycling, and sustaining ecosystem stability (Cui et al., 2022; Wohl et al., 2019).

Most existing studies have focused on hydrological connectivity at large spatial scales, such as watersheds or river basins, with particular emphasis on longitudinal and lateral connectivity (Lizaga et al., 2018; Liu et al., 2021; Flores et al., 2022; Leibowitz et al., 2023). However, research on vertical hydrological connectivity at smaller spatial scales, such as plots, remains limited. Small-scale connectivity plays a fundamental role in shaping large-scale hydrological processes and serves as a basis for understanding broader hydrological connectivity patterns

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(Koestel et al., 2013; Wu et al., 2021). A comprehensive investigation of vertical hydrological connectivity and its controlling factors is essential for elucidating hydrological processes and providing a scientific foundation for effective water resource management at the watershed scale. While hydrological connectivity has been widely examined at watershed scales, its vertical component at pedon scales remains underexplored, despite its critical role in soil-water dynamics.

Vertical hydrological connectivity (VHC), described the ease of water moves from the soil surface to deep layer (Merenlender and Matella, 2013; Cui et al., 2022), is closely associated with preferential water flow— a phenomenon in which a significant portion of water transport within the unsaturated zone occurs through localized pathways, effectively bypassing the surrounding soil matrix with minimal resistance (Beven and Germann, 2013). Soils with well-developed preferential pathways facilitate rapid water infiltration into deeper layers (De Boever et al., 2016; Wang et al., 2016). Given that these pathways facilitate water movement with minimal resistance, preferential flow is a key factor in understanding hydrological connectivity (Bogner et al., 2010; Brevik et al., 2015). Several indicators are commonly used to assess vertical hydrological connectivity, including dye coverage, maximum dyed depth, percentage of preferential pathways, depth of diffusion areas, and length index (Laine-Kaulio et al., 2015; Dai et al., 2020; Cui et al., 2022; Yang et al., 2023). These methods have been widely applied across various ecosystems, such as wetlands (Wu et al., 2021; Dai et al., 2021; Zhang et al., 2021; Cui et al., 2022), riparian zones (Yang et al., 2023), and forest (Zhang et al., 2022; Tang et al., 2024).

To capture flow pathway within soils, various approaches have been proposed to quantify vertical hydrological connectivity, broadly categorized into indirect and direct methods (Allaire et al., 2009). Soil water content inversion is the most widely used indirect approaches, reflecting VHC responses to rainfall; however, it requires long-term monitoring, making it labor-intensive (Bracken et al., 2013; Liu et al., 2019). Among direct approaches, advanced techniques such as computer-based imaging and nuclear magnetic resonance can accurately characterize three-dimensional soil structures (Allaire et al., 2009; Conte et al., 2024). However, these methods are limited to small-scale soil column studies and expensive (Seyfarth et al., 2012; Conte et al., 2024). Comparatively, dye-tracer offers a simpler, more intuitive, and cost-effective alternative (Lipsius and Mooney, 2006; Wang and Zhang, 2011; Dai et al., 2020; Cui et al., 2022). This technique effectively distinguishes hydrological processes and visualizes water flow pathways within soil profiles (Zhang et al., 2016). Therefore, dye-tracer technology provides a practical and efficient approach for quantifying soil vertical hydrological connectivity.

With global climate change and the increasing pressure on the environment, such as soil degradation, land use conversion has emerged as a critical strategy for environmental and resource management worldwide (Nedd et al., 2021). Several studies have examined hydrological connectivity in response to land use changes at large spatial scales (López-Vicente et al., 2013; Lizaga et al., 2018). For instance, Lizaga et al. (2018) found that a reduction in agricultural land led to decreased hydrological connectivity in a Mediterranean catchment, while López-Vicente et al. (2013) reported that increasing vegetation cover reduced hydrological connectivity in the Spanish Pre-Pyrenees. However, large-scale studies primarily identify general connectivity patterns but often lack mechanistic insights into underlying processes. In contrast, small-scale investigations of vertical hydrological connectivity (VHC) can reveal the mechanisms governing hydrological processes. At small scale, land use changes alter vegetation cover, soil structure, and nutrient dynamics, thereby influencing water movement and VHC (Buendia et al., 2016; Jacka et al., 2021; Ding et al., 2023; Jiang et al., 2023; Qiu et al., 2023). Qiu et al. (2023) demonstrated that farmland exhibited the weakest VHC, characterized by the lowest dye coverage area, maximum dyed depth, and depth of diffusion, followed by grassland, shrubs, and forest. Similarly, alder stands exhibited stronger VHC, with a greater maximum dyed depth than oak and spruce stands (Jacka et al., 2021). Jiang et al. (2023) found that dye coverage, an indicator of VHC, followed a decreasing trend across different land uses: tropical rainforest, rubber rainforest, rubber-tea agroforestry, and rubber monoculture.

Previous research has shown that dye coverage is negatively correlated with bulk density and microaggregate stability but positively correlated with soil sand content, organic matter content, root biomass, and root morphological traits (Dai et al., 2021; Cui et al., 2022; Wei et al., 2024). Paddy soils, typically characterized by high clay and organic matter content, relatively high bulk density, and low porosity, significantly influence vertical water movement (Yi et al., 2020). When agricultural land is converted to orchards, the increased soil sand content and extensive root systems in orchard ecosystems enhance infiltration and hydrological connectivity (Liao et al., 2023). Moreover, the effects of past land use on soil physical, hydraulic, and nutrient properties can persist for decades (Foster et al., 2003; Bonell et al., 2010; Ouyang et al., 2013; Lozano-Baez et al., 2019), potentially leading to long-term variations in VHC. Despite these insights, research on VHC under different land use conversion patterns remains limited. While the influence of soil intrinsic properties, such as bulk density and texture, and plant roots on VHC has been widely acknowledged, the relative contributions of these factors remain unclear. Further research is needed to quantify their respective roles in shaping VHC across various land use types.

This study uniquely quantifies VHC under distinct land use conversion types, identifying its key controlling factors and implications for orchard water management, a perspective often overlooked in prior research. Specifically, we examined how alterations in soil profile structure and root distribution, both significantly influenced by land use changes, regulate vertical water movement. By integrating soil properties, root parameters, and hydrological connectivity indicators, this research provides novel insights into the mechanisms driving variations in VHC under different land use patterns. To evaluate these impacts, dye-tracer experiments were conducted in the Three Gorges Reservoir area, a region that has experienced extensive land use conversion over the past decades in an effort to balance ecological preservation with economic development. Due to variations in field management practices, paddy fields and drylands exhibit significant differences in organic matter content, soil nutrients, structure, and stability. These properties are expected to undergo substantial modifications following their conversion into orchards, potentially affecting VHC and regional hydrological processes. The primary land use conversions investigated in this study involve the transformation of paddy fields (P-O sites) and drylands (D-O sites) into citrus orchards. To ensure comparability, orchards with similar citrus tree age and terrain characteristics were selected, while other conditions, including field management practices and soil parent material, were kept consistent throughout the study. We hypothesize that variations in VHC are induced by differences in land use conversion types. Accordingly, the specific objectives of this study were: (1) to assess the effects of various land use conversions on VHC indicators and their potential influencing factors; and (2) to explore the underlying mechanisms through which land use conversions influence VHC. The findings of this research will enhance the understanding of soil hydrological processes following the conversion of croplands into orchards and provide a foundation for optimizing land management strategies to improve water retention, nutrient cycling, and ecosystem sustainability.

2. Materials and method

2.1. Study area

This study was conducted in the Wangjiaqiao watershed (110°40′E-110°47′E, 31°05′N-31°15′N) in Hubei Province, China (Fig. 1), covering 16.7 km^2. The watershed has a continental monsoon climate with mean annual precipitation of 1016 mm. Approximately 70 % of the total

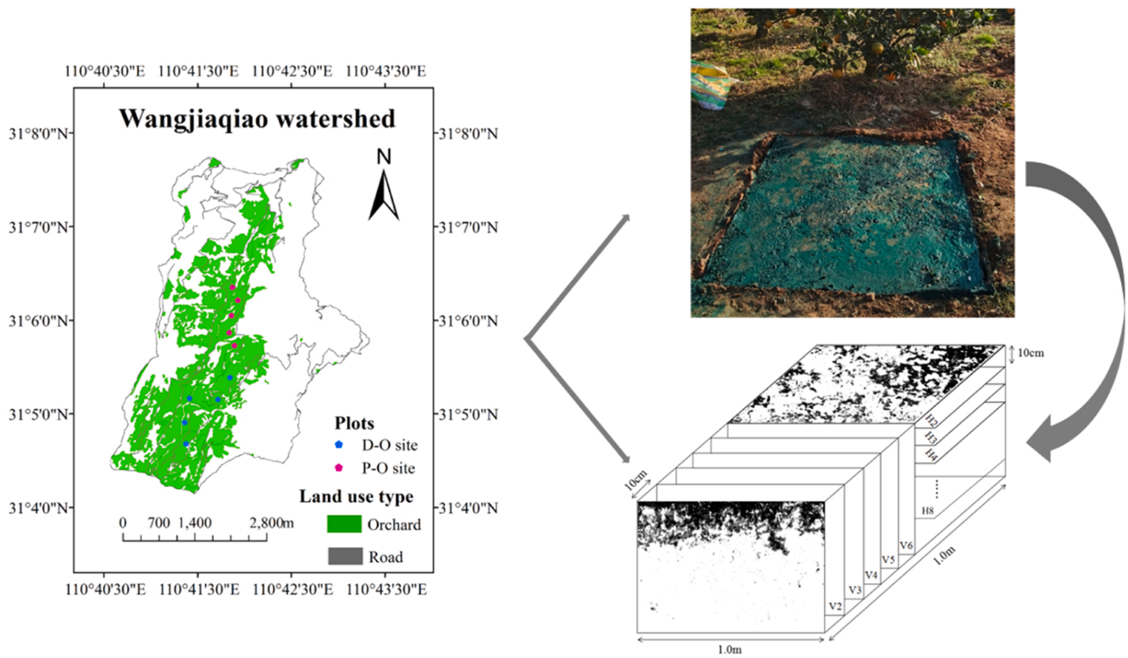

Fig. 1. Location of the study area and process of experiment.

annual precipitation falls between May and September. Elevations within the watershed range from 184 to 1180 m. The watershed is predominantly covered with paddy soils and purple soils. In alignment with policies advocating for the reversion of farmland to forest and the pursuit of economic gains, a notable portion of cultivable land within the watershed, comprising both drylands and paddy fields, has been converted into orchards over the past few decades. The orchard area of the watershed area increased from only 4.43 % in 1995 to 37.44 % in 2015 (Zhou et al., 2023). The per capita income generated from orchard production accounted for 78.69 % of the total per capita income (Deng et al., 2023). This watershed is a typical demonstration area for intensive orchards. This land use conversion resulted in simplified landscapes, which significantly changed hydrological process, such as reducing runoff and sediment load in the watershed (Zhou et al., 2023).

In this study, orchards converted from paddy fields (P-O sites) and drylands (D-O sites) were selected. These orchards distributed in areas with similar soil types and terrain characteristics. The orchards were primarily composed of fifteen-year-old citrus, with the similar field management including no ploughing and irrigation, weeding, and fertilization regime.

2.2. Soil collection and field experiments

Soil collection and field experiments were conducted in five replication plots within each type of orchard. The distance between each plot was more than 3 m (Dai et al., 2021). To ensure stable results, this study was conducted after a minimum of two weeks without precipitation (Dai et al., 2020). A quadrat measuring 1.2 m × 1.2 m was established at a distance of 0.5 m from the tree trunk in the center of plot (Cai et al., 2024). To ensure minimal impact of slope on solution infiltration, the steel frame was placed into the soil in a level area. All litter and weeds were meticulously cleared to reduce surface interference. Soil sample were collected from five points with the sampling depths of 0–10 cm, 10–20 cm, 20–40 cm, 40–60 cm, and 60–80 cm near the quadrat. Because the roots of citrus trees are mainly concentrated within 0–60 cm soil layer, and only a few roots extend to 80 cm. Additionally, the maximum depth of water infiltration was observed to be 79 cm in the actual dye-tracer experiment in this study. Soil physicochemical properties and measurements were summarized in Table S1. In addition, stratification ratio (SR), calculated by Eq. (1), was adopted to indicate the profile distribution pattern of soil properties and root parameters (Fan et al., 2024).

$$SR = \frac{I_{topsoil}}{I_{subsoil}} \quad (1)$$

where, $I_{topsoil}$ indicates the value of soil properties and root parameters in the topsoil (0–10 cm), $I_{subsoil}$ indicates the value in the subsoil (10–20, 20–40,40–60,60–80 cm). The SR > 1 represents the properties decreased with soil depths, SR < 1 represents the properties increased with soil depths. The closer the SR is to 1 represents the properties is more uniform along profile.

Vertical hydrological connectivity was determined by dye-tracing experiment (Dai et al., 2020; 2021; Zhang et al., 2022b; Cui et al., 2022). During in this procedure, a total volume of 72 L of dye solution (C.I. Food Blue 2 with a concentration of 5 g/L) was slowly and uniformly sprayed across the surface of each quadrat using an electric knapsack sprayer, and the application was paused when necessary to prevent ponding or runoff. After the dye application was complete, plastic film was utilized to cover the quadrat, preventing input from rainfall and surface evaporation. After a 24-hour period, to mitigate boundary effects, the central area measuring 1.0 m × 1.0 m within the quadrat was selected as the sampling region. Herein, soil samples were excavated and collected from both the vertical and horizontal profiles. Six vertical profiles were excavated with intervals of 10 cm between slices. After each profile was prepared, dye patterns were photographed with a Canon EOS 80D digital camera, which has a resolution of 5742 × 3648 pixels. The excavation depth was determined by the depth of the dye staining. The photographs were then transformed into grayscale images and binarized to obtain black-and-white biphasic pictures using Adobe Photoshop CC 2017. The binary images were digitized by using the "Bitmap Analysis" function in Image-Pro Plus 6.0, and a numerical matrix containing only two values (0 and 255) was obtained. Subsequently, the feature value of the dyed image was calculated using Excel 2013. The dye indexes in the soil profile served as indicators of hydrological connectivity within the soil (Table 1).

Eight horizontal profiles were also separated with intervals of 10 cm between slices because the maximum depth of dye coverage was 79 cm. Each layer of the horizontal profile was evenly divided into 8 equal

Table 1
The vertical hydrological connectivity indicators.

Indexes	Calculate equation	
Dye coverage (*DC*)	$DC = \frac{D}{D+ND} \times 100\%$ Where, *DC* is the total dye coverage of each profile (%), *D* represents the area that is dyed (cm^2), and *ND* represents the area that is undyed (cm^2).	Any change in the dye coverage within the profile reflects a change in the movement of water, indicating a change in the strength of small-scale hydrological connectivity.
Maximum dyed depth (MDD)	–	It is the depth to which the dye could reach the deepest (cm).
Length index (L_i)	$L_i = \sum_{i=1}^{n} \lvert D_{c(i+1)} - D_{ci} \rvert$ Here, D_{ci}(%) represents the dye coverage of layer I within the profile while $D_{c(i+1)}$ (%) represents that of layer i + 1; and n is the total number of layers, which is six in this paper.	The length index (L_i %) is the sum of the absolute values of the difference between every two adjacent layers within each vertical profile.
Depth of diffusion area (U_{niFr})	–	The depth of diffusion area (U_{niFr},cm) indicates that when the dye coverage is reduced to 80 %, the predominant water flow movement is diffusion. This depth of water flow is referred to as the depth of diffusion area.
Percentage of preferential pathway (P_{F-fr})	$P_{F-fr} = \frac{1 - 100 * U_{niFr}}{S_{Tot}} \times 100\%$ Where P_{F-fr} is the percentage of preferential pathway (%), U_{niFr} is the depth of the diffusion area (cm), S_{Tot} is the total area in the image (cm^2), and 100 indicates that width of the vertical profile is 100 cm.	The percentage of preferential pathway ($P_{F\text{-}fr}$) is the percentage of the preferential flow area that is dyed, measured against the total dyed area in the image.

parts, and all roots within the soil blocks were collected. The size of each soil block was 25 cm × 25 cm × 10 cm. Care was taken to minimize damage to the root system during washing. The washed roots were then dried at 70°C for 48 hours to measure the root biomass (RB). Subsequently, root parameters, including root length (RL), root surface area (RSA), and root volume (RV), were measured using Root Analysis software.

2.3. Statistical analysis

Two-way ANOVA was conducted to evaluate the effects of soil depth and land use conversion patterns on soil properties and root parameters. One-way ANOVA was applied to assess the impact of soil depth on the stratification ratio (SR) and the influence of land use conversion patterns on vertical hydrological connectivity indicators, including dye coverage (DC), maximum dyed depth (MDD), length index (L_i), the depth of diffusion area (U_{niFr}), and preferential flow fraction ($P_{F\text{-}fr}$). Spearman correlation analysis was performed to examine the relationships between SR, soil properties, root parameters, and vertical hydrological connectivity indicators. Additionally, variance partitioning analysis (VPA) was utilized to quantify the relative contributions of explanatory variables (SR of soil properties and root parameters) to the variation in response variables (vertical hydrological connectivity indicators). Furthermore, partial least squares path modeling (PLS-PM) was employed to analyze the key factors influencing vertical hydrological connectivity indicators. Spearman correlation, VPA and PLS-PM analyses were executed using the R package “Hmisc”, “vegan”, “plspm”, respectively. Additionally, one-way ANOVA and two-way ANOVA analyses were carried out using SPSS Statistics ver. 26.0 (SPSS Inc., Chicago, IL, USA).

3. Results

3.1. Impacts of land use conversions on soil profile structure and root parameters

Following a comparable duration of land use conversion, significant disparities in soil layer structure were observed between the P-O and D-O sites ($p < 0.05$) (Figs. 2–4; Table S2 and S3). Specifically, the average clay content and capillary porosity (CP) in the P-O sites ranged from 15.42 % to 36.86 % and 33.46–45.40 %, respectively, both of which were higher than those in D-O sites. These findings indicate that the soil texture in P-O sites is clay loamy with relatively compact, exhibiting limited drainage capacity. In contrast, the sand content, gravel content, and non-capillary porosity (NCP) in D-O sites ranged from 28.51 % to 57.43 %, 0.15–0.86 %, and 1.89–9.57 %, respectively, following an opposite trend. These characteristics suggest that the soil texture in D-O sites is sandy loam, with relatively losser and enhanced air permeability and drainage capacity. The differences in the average silt, bulk density (BD), total porosity (TP), soil organic matter (SOM), soil total nitrogen (STN), and soil total phosphorus (STP) contents between the P-O and D-O sites were not particularly remarkable. However, in each soil layer, TP in the P-O sites exceeded that in the D-O sites only within the 0–10 cm depth. SOM, STN, and STP contents in the P-O sites were significantly higher than those in the D-O sites within the 0–20 cm depth; however, these values displayed an opposite trend within the depth of 20–80 cm. Soil BD, clay, and gravel content exhibited an increasing trend with soil depth, whereas TP, CP, sand, SOM, STN, and STP displayed a contrary trend. Furthermore, the stratification ratio (SR) of sand, BD, TP, CP, NCP, and STP exhibited significant variations across soil layers in P-O sites, whereas these differences were not statistically significant in D-O sites. However, the SR of other soil properties demonstrated notable variations across soil layers in both P-O and D-O sites. These findings indicate that soil property variations with depth were more pronounced in P-O sites, leading to more distinct soil stratification compared to D-O sites.

It showed that land use conversion type had significant impacts on root parameters ($p < 0.05$) (Figs. 5 and 6; Table S3 and S4). In total, root biomass density (RBD), root length density (RLD), root surface area density (RSAD), and root volume density (RVD) in P-O sites, were 10.52 ± 2.81 g 100 cm^{-3}, 1066.29 ± 50.13 cm 100 cm^{-3}, 29.59 ± 2.93 cm^2 100 cm^{-3}, 0.12 ± 0.02 cm^3 100 cm^{-3}, respectively, were higher than the values at D-O sites by 27.36 %, 37.59 %, 71.64 % and 83.33 %, respectively (Table S4). In each soil layer, the RBD, RLD, RSAD and RVD in P-O sites were greater than those in D-O sites at a depth of 0–20 cm, but lower than those in D-O sites at a depth of 20–80 cm. The results suggested that the RBD, RLD, RSAD and RVD at orchards decreased across profile (Fig. 5). Root distribution patterns also varied between the two land use types. In P-O sites, most roots were concentrated within the 0–20 cm depth, with root biomass density (RBD), root length density (RLD), root surface area density (RSAD), and root volume density (RVD) accounting for 85.03 %, 93.28 %, 91.84 %, and 93.26 % of the total root system, respectively. In contrast, the proportion of these root parameters within the 0–20 cm depth in D-O sites was lower, at 70.80 %, 79.83 %, 73.19 %, and 78.57 %, respectively. These results suggested that although total root biomass was higher in P-O sites, roots were predominantly distributed in the surface soil layer, whereas D-O sites exhibited a more developed and efficient deep root system. The SR and its CV values of RBD, RLD, RSAD, and RVD between each soil layers in P-O sites were higher than those values in D-O site, indicating the root distributed more uniformly along profile at D-O site compared with P-O site (Table S3; Fig. 6).

3.2. Impacts of land use conversions on vertical hydrological connectivity indicators

It showed that land use conversion types had significant impacts on

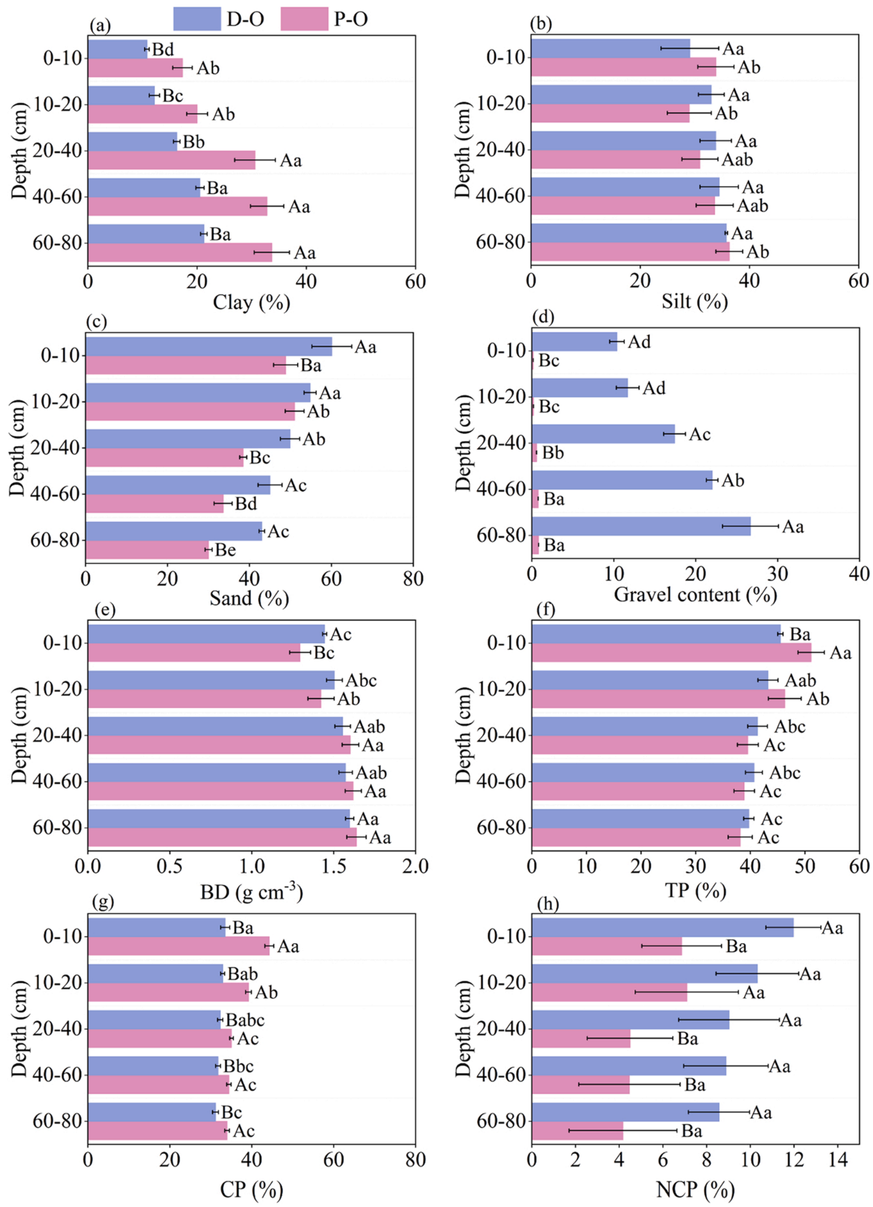

Fig. 2. Variation of soil physical properties with soil depths at P-O and D-O sites. Note, Bars with different lowercase letters denote significant differences among soil depth at $p < 0.05$, and different uppercase letters indicate significant differences between P-O sites and D-O sites. BD, Bulk density; TP, Total porosity; CP, Capillary porosity; NCP, Non-capillary porosity.

vertical hydrological connectivity ($p < 0.05$) (Table 2; Fig. 7 and S1). The DC and U_{niFr} in P-O sites, with a range of 14.52–15.75 % and 0.64–0.69 cm, respectively, were lower than the values in D-O sites. While the MMD, L_i and P_{F-fr}, with a range of 72.50–77.33 cm, 18.26–19.80 and 99.31–99.36 % showed opposite trend, but the differences of MMD, L_i was insignificant. Lower DC and U_{niFr} values in P-O sites indicate a smaller dyed area and higher heterogeneity, suggesting that vertical hydrological connectivity in P-O sites was weaker than in D-O sites. In term of soil profile (Fig. 7 and S1), the DC at P-O sites showed a significant trend of initially decreasing, then increasing, and finally decreasing with increasing soil depth, peaking at 49.26 % at 10–20 cm, indicating noticeable lateral seepage. For the D-O sites, the DC gradually decreased with increasing soil depth. In the surface soil layer (0–20 cm), the DC in the P-O sites increased from 45.37 % to 49.26 %, while that in D-O sites declined from 83.78 % to 55.88 %. For subsurface soil layer (20–80 cm), the DC at P-O and D-O sites varied from 49.26 % to 1.37 % and 55.88–2.29 %, respectively. This result suggested that the vertical hydrological connectivity was much stronger during surface soil layer, especially for D-O sites.

3.3. Determinants to vertical hydrological connectivity indicators

Correlation analysis revealed that DC, MMD, and U_{niFr} were significantly positively correlated with the SR of BD and GC but negatively correlated with the SR of sand, TP, NCP, SOM, STN, STP, RBD, RLD, and RSAD. These findings suggest that a higher stratification ratio of BD and GC, combined with lower SR values for sand, porosity, nutrients, and root parameters, resulted in smaller differences in these properties between soil layers, which ultimately promoted vertical hydrological connectivity. L_i showed statistically positive correlations with the SR of CP (Fig. 8). The SR of soil properties explained the majority of the variation in vertical hydrological connectivity indicators (86 %), whereas the SR of root parameters explained 16 % of the variation (no unique effect) (Fig. 9). The path model was developed to clarify the intricate relationships of the SR of soil properties and root parameters with vertical hydrological connectivity indicator (Fig. 10). The good-of-

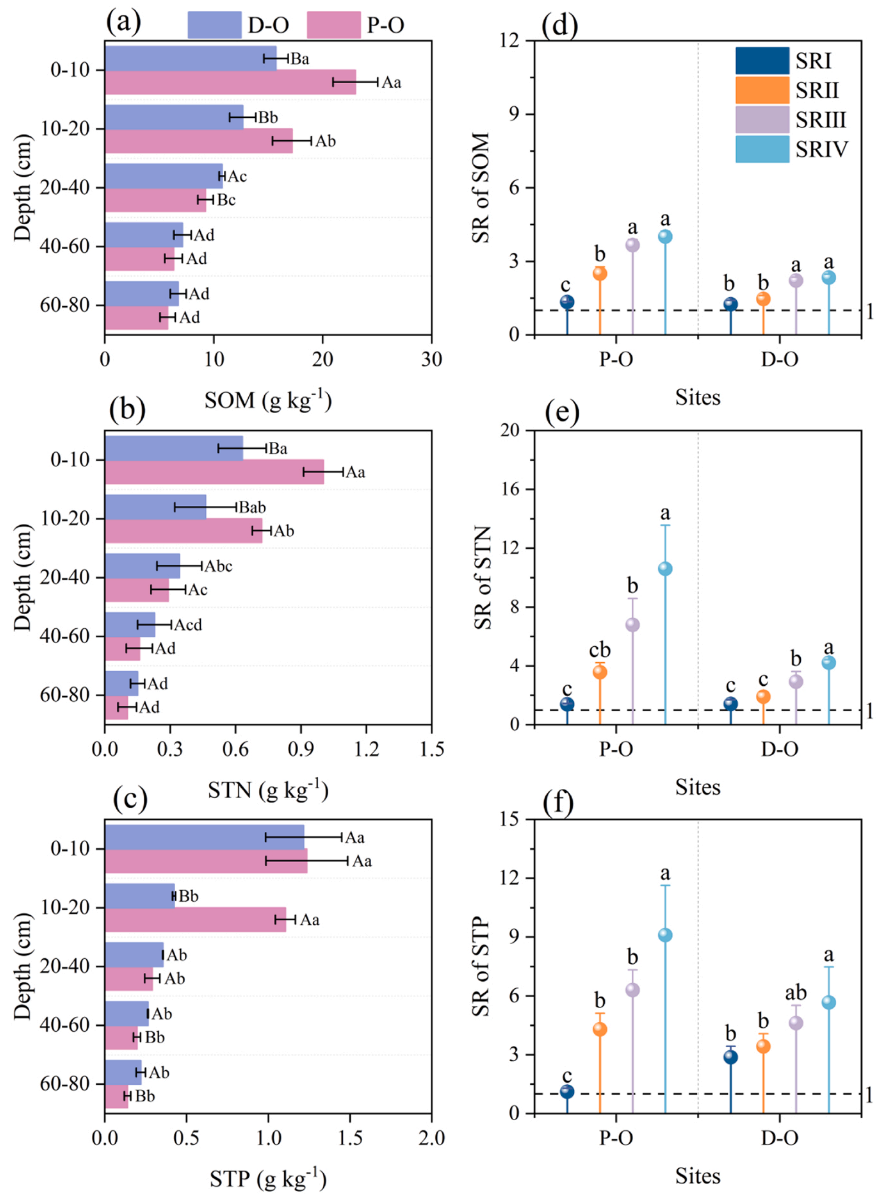

Fig. 3. Variation of soil nutrient contents (a, b, c) and their stratification ratio (SR) (d, e, f) with soil depths at P-O and D-O sites. Note: SRI, SRII, SRIII, SRIV represents the ratio of properties between 0 – 10 and 10–20, 20–40, 40–60, 60–80 cm. SOM, Soil organic matter; STN, Soil total nitrogen; STP, Soil total phosphorus.

fit (GoF) of this model was 0.80. The model explained 49.7 % of the variance in DC and U_{niFr}. Land use conversion influenced vertical hydrological connectivity through affecting the SR of soil properties and root parameters. The SR of soil porosity (TP and NCP) directly and significantly affected DC and U_{niFr} with the SPC of total effect was −0.60. Although SR of soil structure (sand and gravel content), soil nutrients (SOM, STN, and STP) and root parameters (RBD, RLD, and RSAD) had no direct effect on DC and U_{niFr}, they still exhibited evident total effects on DC and U_{niFr} with the SPC of 0.80, 0.55 and −0.39, respectively. These results revealed that land use conversion patterns significantly influenced the profile distribution characteristics of soil texture and nutrients. These variations regulated the vertical distribution of plant roots, thereby influencing soil pore structure, particularly non-capillary porosity. Ultimately, these changes resulted in significant differences in soil hydrological connectivity among different land use patterns.

4. Discussion

Hydrological connectivity, especially the lateral hydrological connectivity, as one of the critical factors to soil ecological and hydrological functions, has been widely investigated at watershed and/or regional scales (Liu et al., 2021; Flores et al., 2022; Leibowitz et al., 2023). Due to the limited information on vertical hydrological connectivity at small scales, such as plots and pedons, elaborated simulations of soil hydrological process still remain challenging (Wu et al., 2021). In this study, vertical hydrological connectivity of soils in orchards converted from paddy fields (P-O sites) and drylands (D-O sites) was determined by dye tracer experiments. Although dye tracer techniques have certain limitations in characterizing vertical hydrological connectivity, such as the influence of pre-existing soil moisture on the dye solution and the inability of dying patterns to fully capture actual hydrological connectivity. Additionally, high temperatures during experiment can cause dye evaporation before reaching deeper soil layers, potentially underestimating the dyed area. However, dyed areas indicate interconnected flow path. When combined with soil indicators, this dye tracer technique

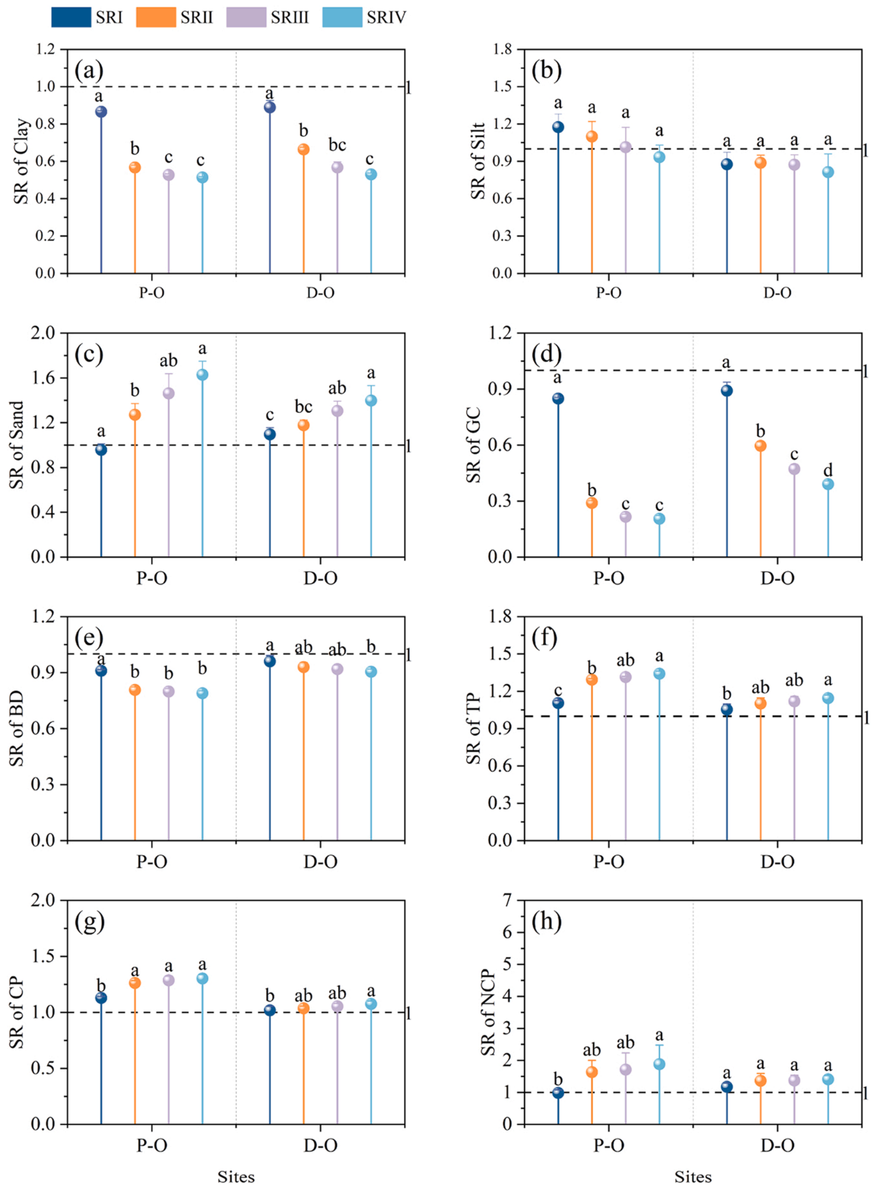

Fig. 4. The stratification ratio (SR) of soil physical properties with soil depths at P-O and D-O sites. Note: SRI, SRII, SRIII, SRIV represents the ratio of properties between 0 – 10 and 10–20, 20–40, 40–60, 60–80 cm. GC, Gravel content; BD, Bulk density; TP, Total porosity; CP, Capillary porosity; NCP, Non-capillary porosity.

provides rapid and intuitive insights into the spatial and temporal distribution of vertical hydrological connectivity (Dai et al., 2020). Furthermore, our experiments were conducted after two consecutive weeks without rainfall, ensuring stable initial soil moisture conditions. The winter timing of our study, with its low temperatures, effectively minimized evaporation effects. These controlled conditions enhanced the accuracy and reliability of our findings.

Compared to the P-O sites, the D-O sites exhibited significantly higher sand content, non-capillary porosity (NCP), and gravel content (GC), along with a lower stratification ratio (SR) of soil properties (Figs. 2 and 4, Table 2). These differences are attributed to the distinct soil properties in the Three Gorges Reservoir Region, where paddy soils typically contain higher clay content, whereas dryland soils are characterized by greater sand content, NCP, and GC (Dai et al., 2022; Li et al., 2022). The findings indicate that soil properties in orchards are influenced by land use history. Previous studies have consistently demonstrated the long-term effects of historical land use on soil characteristics. For instance, Sonneveld et al. (2003) reported that pastures reseeded from abandoned fields exhibited significantly higher clay content, organic matter, saturated water content, and hydraulic conductivity than those converted from maize-cultivated land, with lower bulk density. Similarly, Zimmermann et al. (2006) found that the impact of teak plantation afforestation on soil hydraulic properties strongly depended on previous land use. These studies collectively highlight that the legacy effects of historical land use can persist for decades, significantly shaping soil properties and hydrological processes.

The DC, MMD, and U_{niFr} values at the D-O sites were higher than those at the P-O sites, whereas $P_{F\text{-}fr}$ was lower (Table 2), indicating stronger vertical hydrological connectivity at the D-O sites (Dai et al., 2020). The SR of sand was significantly negatively correlated with DC and U_{niFr} (Fig. 8), suggesting that a smaller decrease in sand content from topsoil to subsoil enhanced vertical water flow and improved hydrological connectivity (Marquart et al., 2020; Zhu et al., 2020; Qiu et al., 2023). This is attributed to the high sand content derived from purple sandstone parent material, which has lower viscosity and soil water suction compared to clay and silt, facilitating more efficient water movement (Zhao et al., 2022). Previous studies have shown that coarse-textured soils with high sand content promote more

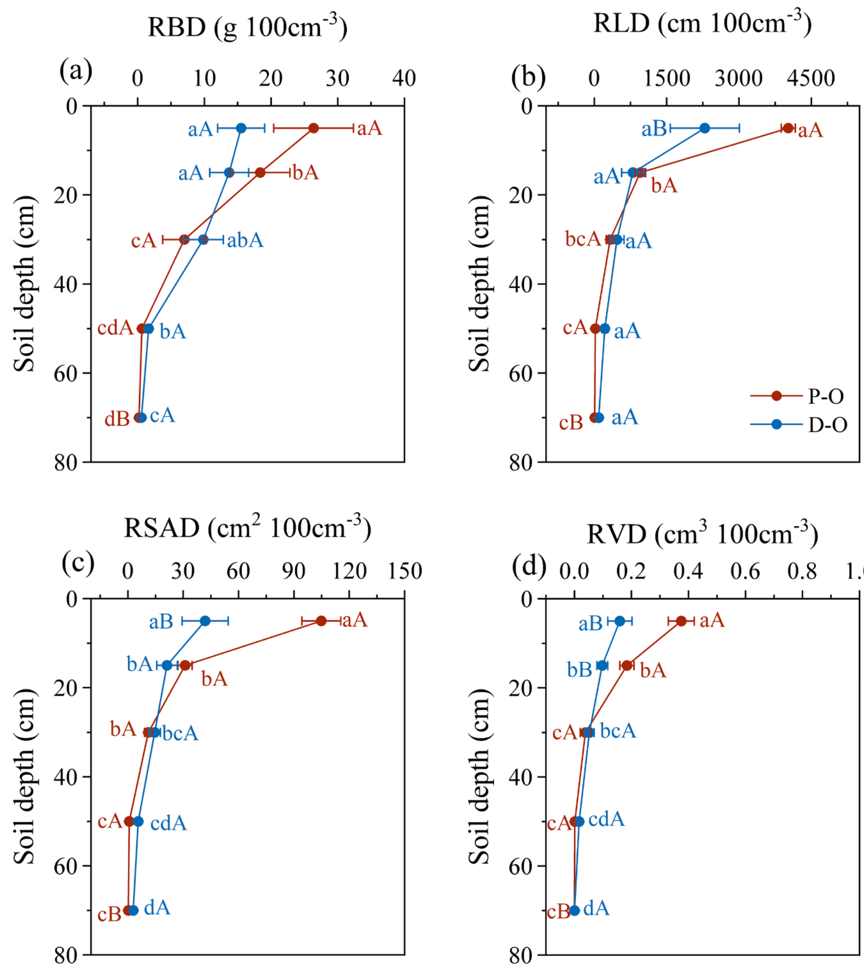

Fig. 5. Variation of root parameters with soil depths at P-O and D-O sites. RBD, Root biomass density, RLD, Root length density, RSAD, Root surface area density, RVD, Root volume density.

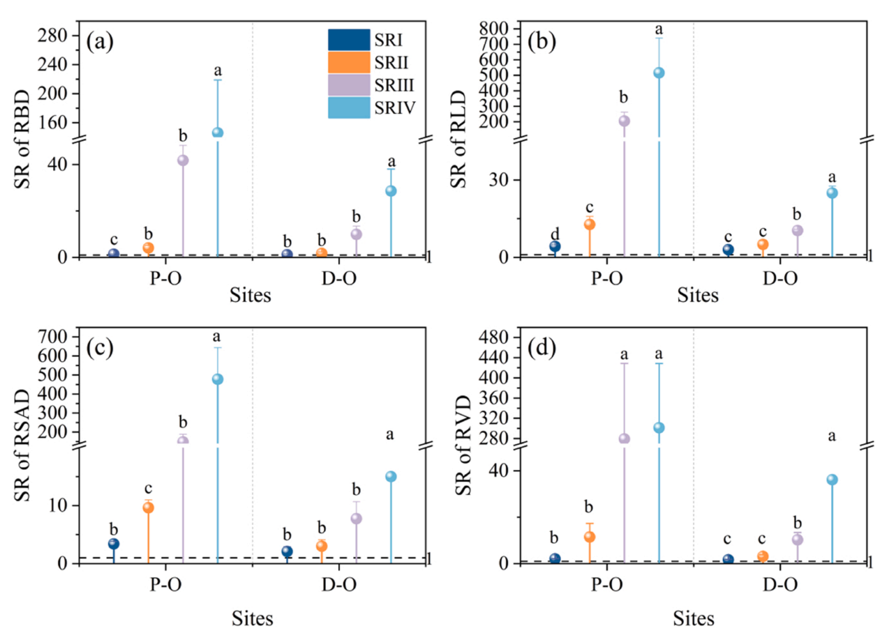

Fig. 6. The stratification ratio (SR) of root parameters with soil depths in P-O and D-O sites. RBD, Root biomass density, RLD, Root length density, RSAD, Root surface area density, RVD, Root volume density.

Table 2
Vertical hydrological connectivity indicators of soil profiles in the field plots.

Sites	P-O	D-O
DC (%)	15.14 ± 0.62B	24.81 ± 0.26 A
MDD (cm)	75.33 ± 1.92 A	78.17 ± 1.15 A
L_i	19.03 ± 0.77 A	18.86 ± 0.19 A
U_{niFr}(cm)	0.66 ± 0.03B	8.31 ± 0.09 A
$P_{F\text{-}fr}$ (%)	99.34 ± 0.03 A	91.69 ± 0.09B

Note: Different letters indicate significant different between treatments at $p < 0.05$ level. DC, Dye coverage; MMD, Maximum dyed depth; L_i, Length index; U_{niFr}, Depth of diffusion area; $P_{F\text{-}fr}$, Percentage of preferential pathway.

homogeneous water infiltration, enhancing hydrological connectivity (Yao and Hendrickx, 1996). Additionally, the significant negative correlation between the SR of NCP and both DC and U_{niFr} suggests that a slower decline in NCP with depth promotes hydrological connectivity (Luo et al., 2023). The higher NCP observed at the D-O sites thus contributed to stronger vertical hydrological connectivity. These findings align with prior research emphasizing the critical role of macropores in sustaining vertical water movement and connectivity (Flury et al., 1994; Beven and Germann, 2013; Jarvis et al., 2016). Furthermore, Zhang et al. (2024) identified a significant positive correlation between soil porosity (TP and NCP) and DC, confirming the strong influence of NCP on vertical hydrological connectivity. The SR of GC was

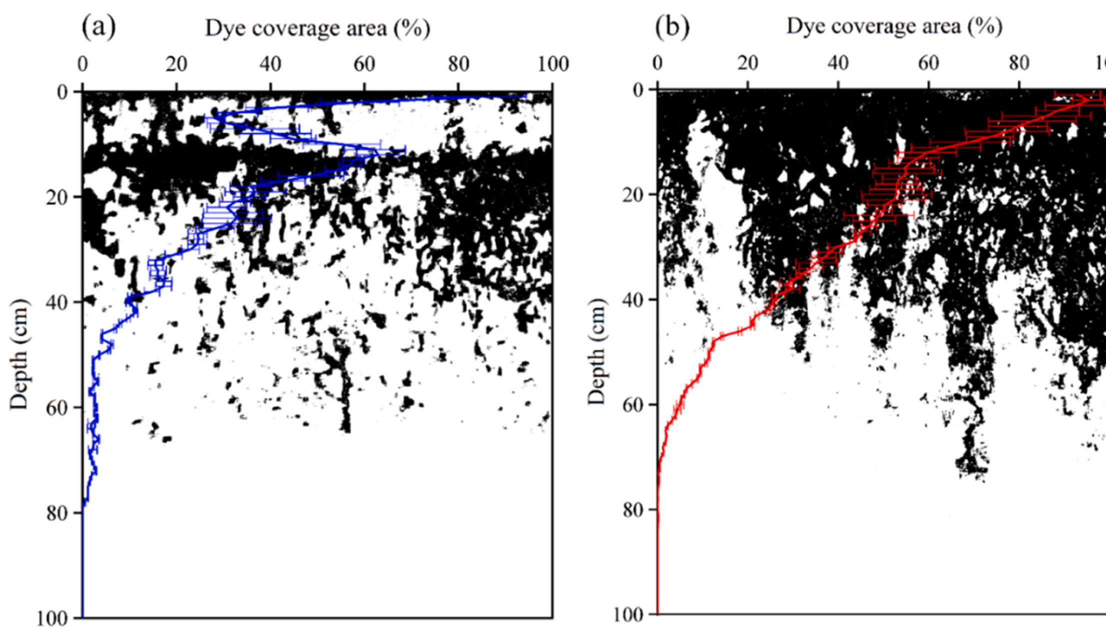

Fig. 7. The change of vertical hydrological connectivity indicator (dye coverage) along soil profile in P-O (a) and D-O sites (b).

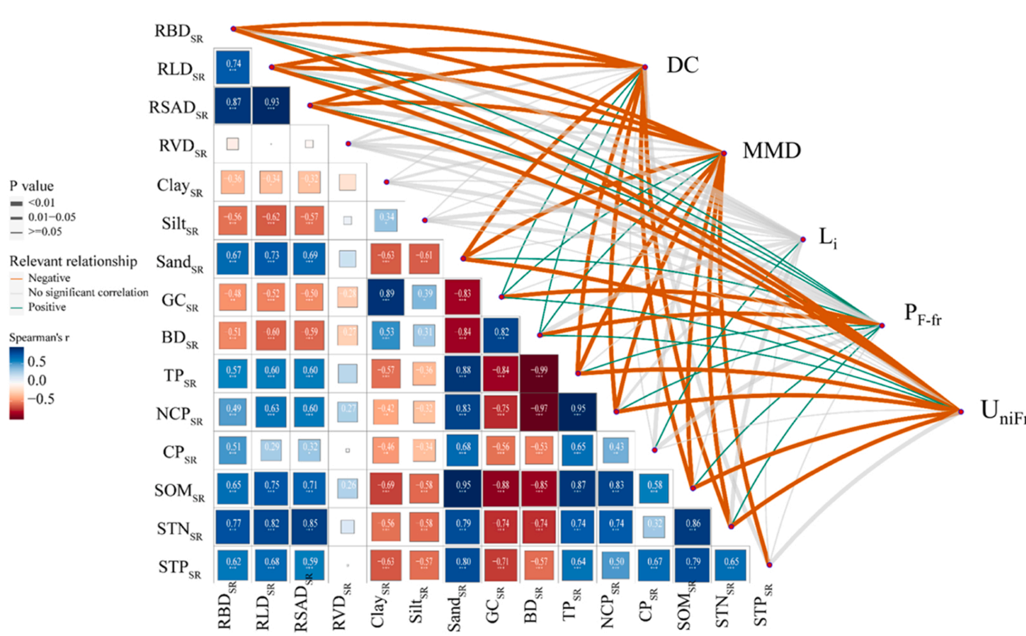

Fig. 8. Spearman's correlation of stratification ratio (SR) of soil properties, root parameters with vertical hydrological connectivity indicators. RBD, Root biomass density, RLD, Root length density, RSAD, Root surface area density, RVD, Root volume density; GC, Gravel content; BD, Bulk density; TP, Total porosity; CP, Capillary porosity; NCP, Non-capillary porosity; RBD, Root biomass density, RLD, Root length density, RSAD, Root surface area density, RVD, Root volume density. DC, Dye coverage; MMD, Maximum dyed depth; L_i, Length index; U_{niFr}, Depth of diffusion area; $P_{F\text{-}fr}$, Percentage of preferential pathway.

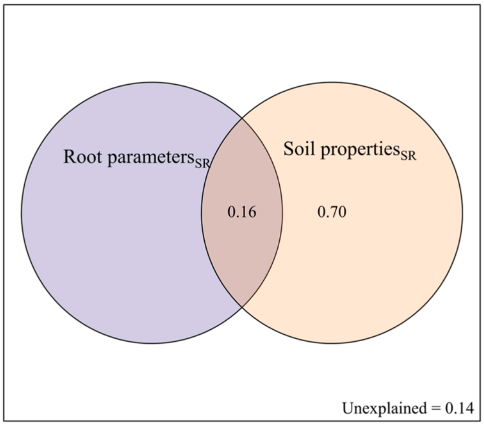

Fig. 9. The effects of stratification ratio (SR) of soil properties (Sand, gravel contents, total porosity, non-capillary porosity, capillary porosity, soil organic matter, soil total nitrogen, total phosphorus) and root parameters (root biomass density (RBD), root length density (RLD), and root surface area density (RSAD)) on vertical hydrological connectivity indicators (dye coverage (DC), maximum dyed depth (MMD), length index (L_i), depth of diffusion area (U_{niFr}), percentage of preferential pathway ($P_{F\text{-}fr}$)).

significantly positively correlated with DC and U_{niFr} (Fig. 8), suggesting that greater variation in gravel content across soil layers enhanced water flow and strengthened vertical hydrological connectivity. The higher gravel content at the D-O sites improved hydrological connectivity by increasing pore spaces between gravel-soil and gravel-gravel interfaces, facilitating vertical water movement (Lv et al., 2019; Lai et al., 2022). Similar findings were reported in Texas, where soils with high limestone gravel content promoted rapid dye infiltration into deeper soil layers (Nobles et al., 2010).

Dye coverage showed distinct trends with soil depth at the D-O and P-O sites (Fig. 7). At the D-O sites, DC exhibited a continuous decreasing trend along the soil profile, with the highest value in the 0–10 cm layer (Fig. 7 and S1), suggesting strong vertical hydrological connectivity throughout the profile. This can be attributed to SR values approaching 1 at the D-O sites (Fig. 4), indicating minimal vertical differentiation in soil structure, which facilitates water movement. Similar trends have been observed in farmland on the North China Piedmont Impingement Plain, orchards in karst regions, and boreal forests on glacial sandy till, where dye coverage consistently decreased with soil depth (Laine-Kaulio et al., 2015; Zhang et al., 2022; Wei et al., 2024). However, at the P-O sites, DC fluctuated along the soil profile, following a "decrease-increase-decrease" pattern, consistent with previous studies (Zhang et al., 2022a; Dai et al., 2020). This variation is likely due to increasing clay content and bulk density with depth, particularly at 20 cm (Fig. 2). The historical use of the P-O sites as a paddy field led to distinct soil stratification. As soil texture transitioned from loose to compact, the dye solution encountered denser layers, forming hydrological barriers that reduced connectivity (Chelli et al., 2018; Hardie et al., 2018). These barriers significantly retained the dye solution, causing surface accumulation and fluctuations in dye coverage, characterized by sudden increases followed by declines in hydrological connectivity (Zhang et al., 2015; Chelli et al., 2018).

Plant roots were another crucial factor influencing vertical hydrological connectivity, as they are affected by various soil properties, including texture, bulk density, and nutrient availability (Zhao et al., 2023; Cai et al., 2024). Compared to D-O sites, the root biomass density (RBD), root length density (RLD), root surface area density (RSAD), and root volume density (RVD) at P-O sites were higher in the 0–20 cm layer but lower in the 20–80 cm layer (Fig. 5). This suggests that the P-O sites had greater root abundance and a stronger capacity for nutrient and water uptake in the topsoil (Sha et al., 2024). However, root distribution followed the opposite trend in deeper layers, likely due to a hydrological barrier in the orchard converted from a paddy field (Li and Jia, 2022). The higher bulk density and clay content in this barrier layer restricted root penetration, concentrating roots in the upper 20 cm. Additionally, lower subsoil organic matter, total nitrogen, and total phosphorus contents further limited root growth (Pelissier et al., 2021). In contrast, the D-O sites exhibited relatively lower bulk density and clay content in the subsoil, resulting in smaller nutrient decrease rates with depth and allowing citrus roots to distribute more evenly (He et al., 2024). Our results also demonstrated that the SR of root parameters was significantly positively correlated with the SR of soil organic matter and nutrients but negatively correlated with the SR of bulk density and clay content (Fig. 8).

The SR of RBD, RLD, and RSAD was significantly negatively correlated with DC and U_{niFr}, indicating that a slower decrease in root parameters along the soil profile promoted vertical hydrological connectivity (Fig. 8), consistent with findings by Hou et al. (2024) and Huang et al. (2024). Thus, a lower SR of RBD, RLD, and RSAD strengthened vertical hydrological connectivity at D-O sites (Fig. 6).

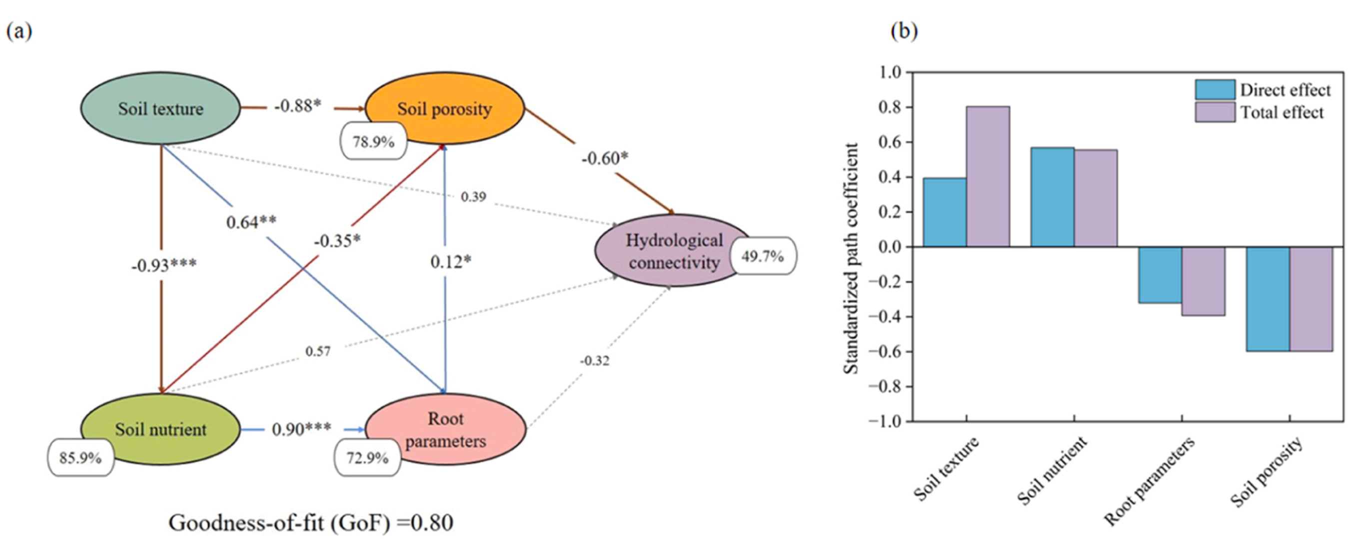

Fig. 10. Effects of land use conversion pattern, stratification ratios of soil texture (sand and gravel contents (GC)), soil nutrient (soil organic matter (SOM), soil total nitrogen (STN), and soil total phosphorous (STP)), plant root (root biomass density (RBD), root length density (RLD), and root surface area density (RSAD)), and soil porosity (total porosity (TP), non-capillary porosity (NCP)) on vertical hydrological indicators (dye coverage (DC) and depth of diffusion area (U_{niFr})).

However, root parameters indirectly affected vertical hydrological connectivity by influencing soil porosity (Figs. 9 and 10). Citrus roots, predominantly distributed within the 0–60 cm layer, are characterized by long taproots, dense lateral roots, and large diameters. These structural features, along with root-soil interactions, contribute to the formation of widely distributed, structurally stable, and continuous pore networks (Schwärzel et al., 2012). These root-induced macropores act as preferential flow channels, facilitating water infiltration, reducing deep percolation time, and minimizing topsoil evaporation losses (Tobella et al., 2014; Cui et al., 2022). Supporting evidence comes from Devitt and Smith (2002), who found that shrub root systems in arid regions created macropores that enhanced vertical water flow in desert soils. Additionally, increased root density strengthened preferential flow, further enhancing hydrological connectivity. Land use conversion significantly influenced preferential flow pathways by modifying root system development (Tobella et al., 2014). In summary, the conversion of drylands to orchards generally supports gradual root system development due to relatively uniform soil properties across different depths. This promotes deeper root penetration and the formation of macropores, progressively enhancing vertical hydrological connectivity over time. In contrast, the conversion of paddy fields to orchards results in pronounced soil stratification, with a compact barrier layer that restricts root proliferation beyond the 0–20 cm surface layer, thereby limiting deeper hydrological connectivity. However, as orchard trees mature, root turnover through senescence and decomposition may gradually increase soil organic matter content and improve porosity. This natural process has the potential to enhance hydrological connectivity in paddy-to-orchard conversions. Nonetheless, empirical evidence on this long-term transformation remains limited, emphasizing the need for extended monitoring to fully understand these dynamics.

The results indicated that orchards converted from drylands exhibited stronger vertical hydrological connectivity (Fig. 11), which enhanced water infiltration and nutrient transportation, thereby increasing citrus yield. In contrast, orchards converted from paddy fields had a noticeable plough pan (Fig. 11), which restricted the downward transport of water and created barriers to nutrient migration and root penetration, ultimately hindering citrus growth and reducing yield (Zhai et al., 2017; Fenta et al., 2022). Furthermore, the presence of a soil barrier layer in orchards converted from paddy fields significantly restricts vertical water infiltration, thereby enhancing surface runoff generation. This hydrological alteration not only intensifies soil erosion processes but also disturbs the equilibrium of regional hydrological cycling, potentially leading to substantial environmental consequences. To enhance vertical hydrological connectivity in orchards converted from paddy fields, management practices aimed at increasing soil porosity should be implemented. Strategies such as deep plowing or intercropping with deep-rooted crops can help break compacted layers, improve soil structure, and facilitate deeper water infiltration. This study emphasized the importance of land use history in shaping vertical hydrological. Therefore, regional hydrological process studies should not only focus on current land use but also consider historical land use changes to improve the accuracy of hydrological simulations and better understand water movement dynamics in human-modified landscapes. However, due to the long conversion period, we were unable to obtain data on vertical hydrological connectivity before the land use change. Future research should focus on long-term monitoring to address this gap.

5. Conclusions

In this study, vertical hydrological connectivity of soils in orchards converted from paddy fields (P-O sites) and drylands (D-O sites) was determined by dye tracer experiments. Vertical hydrological connectivity was characterized by dye coverage (DC), maximum dyed depth (MMD), length index (L_i), uniform infiltration depth (U_{niFr}), and percentage of preferential pathway ($P_{F\text{-}fr}$). Despite the less significant difference in MMD and L_i, DC and U_{niFr} at D-O sites were significantly higher than the values at P-O sites, whereas $P_{F\text{-}fr}$ showed an opposite trend. The enhanced hydrological connectivity at D-O sites could be attributed to the soil structure and roots traits, especially the stratification ratio (SR) of soil texture, porosity, root biomass density, root length density, and root area surface area density. Collectively, the differentiation of vertical hydrological connectivity between P-O and D-O sites was intrinsically controlled by soil porosity. This study focused on a comparison of vertical hydrological connectivity under two typical land use conservation patterns (P-O and D-O). The obtained results suggested that conversion from drylands to orchards may facilitate a better water and nutrient movement and economic outcomes. However, due to the extended period since these conversions occurred, obtaining data on vertical hydrological connectivity prior to land use changes is challenging. Future research should incorporate hydrological modeling alongside long-term temporal monitoring to comprehensively evaluate

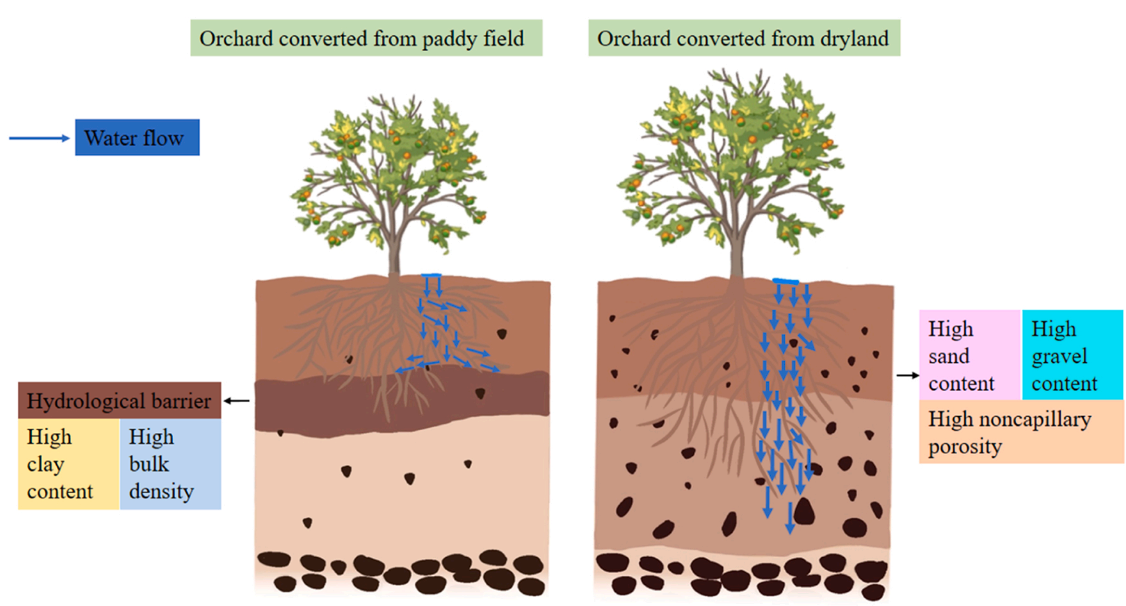

Fig. 11. Schematic diagram of the results.

the dynamic evolution of vertical hydrological connectivity following land-use conversion. This integrated approach would provide deeper insights into the long-term impacts on soil water movement, nutrient cycling, and overall ecosystem sustainability, facilitating the development of more effective land management strategies.

CRediT authorship contribution statement

Wei Yujie: Writing – review & editing, Conceptualization. **Du Yingni:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Li Zhaoxia:** Supervision, Resources, Project administration, Funding acquisition. **Wang Tianwei:** Supervision, Resources, Project administration, Funding acquisition. **Chen Yuwei:** Formal analysis. **Wang Yundong:** Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.still.2025.106631.

Data availability

Data will be made available on request.

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